DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Bangladesh Army University of Science and Technology,Saidpur

PROJECT PROPOSAL REPORT

CSE 3102: Software Engineering Sessional

FarmLink

A Mobile-Based Surplus Food Redistribution and Farm Resource Management System

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Abstract

Every day, university residential halls generate a significant quantity of surplus and leftover food that is discarded as waste. At the same time, nearby livestock farms regularly spend money to procure animal feed. This proposal presents “FarmLink”, a mobile-based platform built with Flutter and Firebase that bridges university halls and local livestock farms through a structured, automated food-redistribution system. Hall administrators post details of surplus food, the system matches and notifies suitable registered farms in real time, and farms collect the food to use as animal feed. In return, farms supply the hall with dairy, eggs, meat, or cash, creating a circular-economy exchange. The system integrates an animal health and nutrition engine, food safety guidelines, QR-based pickup verification, and an analytics dashboard to ensure the process is transparent, safe, and efficient. This report presents the problem background, objectives, proposed system design, feature set, technology stack, feasibility analysis, and work plan for the project, developed as part of the CSE 3102 Software Engineering Sessional course.

1. Introduction

Food waste is one of the most overlooked sustainability problems on university campuses. Residential halls prepare food in bulk, and daily surplus is a near-unavoidable consequence of estimating demand for hundreds of students. This surplus is presently thrown away, contributing to environmental pollution and unnecessary financial loss for the institution. Meanwhile, small and medium livestock farms operating near these campuses continuously face rising feed costs, which directly affects the affordability of dairy, poultry, and meat products for the surrounding community.

FarmLink proposes a technology-driven solution that connects these two otherwise disconnected problems. Rather than treating surplus hall food as waste, the system treats it as an underutilized resource that can be redirected to farms in a controlled, health-conscious, and trackable manner. The result is a mutually beneficial exchange: halls reduce waste and receive food products or compensation in return, while farms reduce feed costs and gain a reliable, recurring feed supply.

2. Problem Statement

- Large volumes of surplus and leftover food from university halls are discarded daily, causing environmental and financial loss.
- No structured or digital mechanism currently exists to connect halls with nearby farms for surplus food redistribution.
- Farm owners lack visibility into what surplus food is available, in what quantity, and when it can be collected.
- Feeding animals unsuitable food types or incorrect quantities can cause health issues, and farm owners often lack standardized nutrition guidance.

- There is no transparent system to track and record the exchange between hall-supplied food and farm-supplied returns (milk, eggs, meat, or cash).
- Manual coordination (phone calls, informal arrangements) is inefficient, unreliable, and difficult to scale or audit.

3. Objectives

The primary objective of this project is to design and develop a cross-platform mobile application that automates and streamlines the redistribution of surplus hall food to registered livestock farms. The specific objectives are:

1. To develop a role-based mobile application (Hall Admin, Farm Owner, Super Admin) using Flutter for both Android and iOS.
2. To build a smart matching and notification system that connects surplus food posts to the most suitable nearby farms in real time.
3. To design an animal health and nutrition engine that guides safe feeding practices across multiple animal types.
4. To implement a food preservation and safety module to reduce contamination risk during transit.
5. To provide a transparent transaction and reward tracking mechanism between halls and farms.
6. To create an analytics dashboard for administrators to monitor food-waste reduction, farm participation, and system-wide impact.
7. To ensure data security and privacy through role-based access control and secure authentication.

4. Proposed System

FarmLink operates as a closed-loop exchange system with three primary actors: the Hall Administrator, the Farm Owner, and a Super Admin who oversees the overall network. The workflow proceeds as follows:

- The hall administrator logs surplus/leftover food details (food type, estimated quantity, and expected pickup time) into the application.
- The system's matching engine identifies eligible, registered farms within a suitable range and sends an automatic push notification via Firebase Cloud Messaging.
- An interested farm owner accepts the request through the app; the Animal Health & Nutrition Engine verifies the food type against the farm's registered animals before confirming the match.
- At pickup, the farm owner scans a QR code generated by the system to confirm collection, which is logged with a timestamp and quantity.
- The farm supplies an agreed return (milk, eggs, meat, fish, or cash) to the hall, which is recorded through the Transaction & Reward module.
- All activity — supply volume, farm participation, and exchange value — is aggregated and visualized on the analytics dashboard for institutional reporting.

This structure converts an informal, wasteful process into a transparent, auditable, and repeatable digital workflow, benefiting the university, the farms, and the surrounding community.

5. System Features and Modules

Module	Description
Smart Supply Matching System	Hall admin posts surplus food data (type, quantity, time). The system automatically matches and notifies the nearest/most suitable registered farm owner.
Animal Health & Nutrition Engine	Built-in guideline and calculator defining safe food types, quantities, and feeding schedules for different animal species (cattle, goat, poultry, fish) to prevent health hazards.
Food Preservation & Safety Module	Provides preservation guidelines and safety checks so surplus food remains uncontaminated between collection and delivery.
Daily Tracking & Analytics Dashboard	Central dashboard tracking daily supply volume, farm production output, and exchange value with visual charts and reports.
Transaction & Reward System	Automatically records the exchange — food supplied by the hall versus milk, eggs, meat, fish, or cash received in return.
QR-based Pickup Verification	Farm owners scan a QR code at pickup to confirm collection, reducing miscommunication and fraud.
Notification System	Real-time push notifications (FCM) and SMS fallback (Twilio) to alert farm owners instantly.
Role-based Access Control	Separate secured interfaces for Hall Admin, Farm Owner, and Super Admin with data isolation via security rules.

6. Proposed Technology Stack

The system will be developed as a cross-platform mobile application supported by a scalable cloud backend, as summarized below:

Layer / Component	Technology
Mobile Application (Android + iOS)	Flutter, Dart
State Management	Riverpod
Backend & Database	Firebase (Firestore)
Authentication	Firebase Auth (role-based: Hall Admin, Farm Owner, Super Admin)
Push Notification	Firebase Cloud Messaging (FCM)
File & Media Storage	Firebase Storage
Server-side Logic	Firebase Cloud Functions
Maps & Location	Google Maps API, Geolocator (Flutter package)
Pickup Verification	QR Code Generation & Scanning (qr_flutter, mobile_scanner)
Offline Local Cache	Hive / SQLite
Admin Web Dashboard	Flutter Web / React.js
Analytics & Reporting	Firebase Analytics, fl_chart
SMS Fallback Notification	Twilio API
Payment / Reward Settlement	bKash / Nagad API
Security	Firebase Security Rules
Version Control	Git & GitHub

Firebase and Supabase are treated as alternative backend choices; the final selection will depend on the relational-data requirements of the transaction and reporting modules identified during the design phase (Supabase/PostgreSQL is preferred if complex relational reporting is required, while Firebase/Firestore offers faster real-time sync for notifications and chat).

7. System Architecture Overview

The system follows a three-tier architecture:

- Presentation Layer: Flutter-based mobile applications for Hall Admin and Farm Owner, plus a web-based dashboard for Super Admin.
- Application/Logic Layer: Firebase Cloud Functions / Supabase Edge Functions handle matching logic, notification triggers, nutrition validation, and transaction calculations.
- Data Layer: Firestore or PostgreSQL stores user profiles, food posts, farm profiles, animal nutrition data, transaction logs, and analytics data, secured through role-based access rules.

Communication between layers is handled through secure REST/real-time APIs, with Firebase Cloud Messaging providing asynchronous, event-driven notifications between hall and farm users.

8. Feasibility Analysis

8.1 Technical Feasibility

The proposed technologies (Flutter, Dart, Firebase/Supabase) are mature, well-documented, and widely used for cross-platform mobile development, making the technical implementation feasible within an academic project timeline.

8.2 Economic Feasibility

Firebase and Supabase both offer free-tier plans suitable for prototype and small-scale deployment, keeping development costs minimal. Optional payment gateway integration can be deferred to a later phase.

8.3 Operational Feasibility

The role-based design (simple posting workflow for hall admins, simple accept/collect workflow for farm owners) minimizes the learning curve, and SMS fallback ensures accessibility for farm owners with limited smartphone literacy.

9. Project Work Plan / Timeline

Phase	Task	Duration
Phase 1	Requirement Analysis, SRS Documentation, ER Diagram & System Design	Week 1 – 2
Phase 2	UI/UX Wireframing (Hall Admin App, Farm Owner App, Web Dashboard)	Week 3
Phase 3	Backend Setup — Database Schema, Auth, Cloud Functions	Week 4 – 5
Phase 4	Flutter App Development — Core Modules (Supply Posting, Matching, Notification)	Week 6 – 8
Phase 5	Animal Health & Nutrition Engine + Food Safety Module	Week 9 – 10
Phase 6	QR Verification, Maps Integration, Analytics Dashboard	Week 11 – 12
Phase 7	Integration Testing, Bug Fixing, Performance Testing	Week 13
Phase 8	User Acceptance Testing, Documentation & Final Presentation	Week 14

10. Expected Outcome and Impact

- Measurable reduction in food waste generated by university halls.
- Reduced feed costs and improved profitability for participating farms.
- Savings in the hall's food budget through returned dairy, eggs, or meat.
- A transparent, auditable record of food redistribution for institutional sustainability reporting.
- A scalable, sustainable circular-economy model that can be extended to multiple university campuses and food-service outlets.

11. Conclusion

FarmLink addresses a real and recurring inefficiency in university food management by converting surplus food from a liability into a shared resource. By combining a role-based mobile application, an automated matching and notification system, and a dedicated animal health and safety engine, the proposed system offers a practical, technically feasible, and socially impactful solution. This project will be developed and evaluated as part of the CSE 3102 Software Engineering Sessional course, following standard software engineering practices including requirement analysis, system design, implementation, and testing.

References

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